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29th RoboRacer Racing @ IFAC 2026 UNICORN Racing Race Stack

High-Assurance Mobility Control Lab.
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Team
UNICORN Racing

UNIST

 **HMCL**
High-assurance Mobility Control Lab

Contents

- **Intro**

- Overall architecture
- Racing philosophy

- **Full stack review**

- Hardware | Compute platform
- Race preparation | Mapping, Raceline optimization
- Localization | Perception
- Planning | Control

- **Result**

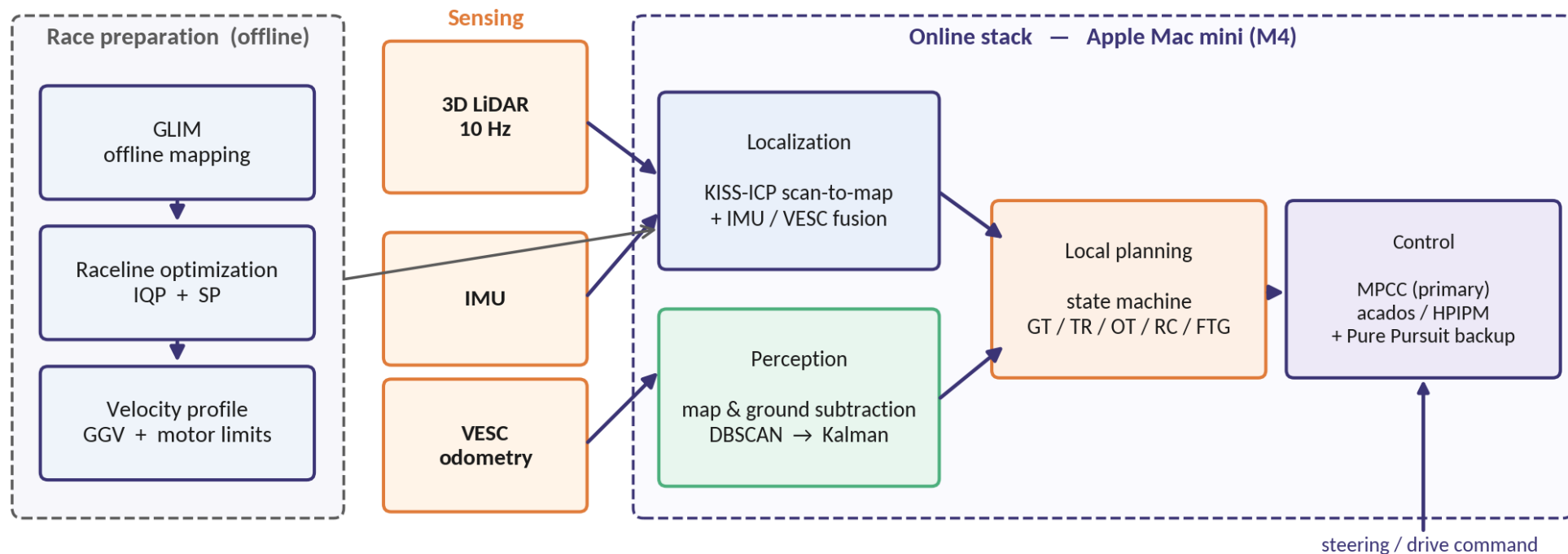
Intro

- Overall architecture
- Racing philosophy

Intro | Overall architecture

Known track, split computation

- The track is known in advance, so we move the heavy computation offline — mapping and raceline optimization — and leave the online stack to localize, perceive, plan and control.
- Every online module runs on a single Apple Mac mini (M4).



Intro | Racing philosophy

1. Zero collisions.

—Tracked obstacles feed corridor- and half-plane-constrained MPCC avoidance.

2. Planned motion, always.

—A state machine selects tracking, trailing, overtaking, recovery, or follow-the-gap.

3. Minimum lap time.

—Offline raceline optimization and GGV profiling provide the speed reference.

4. Optimization over heuristics.

—MPCC trades progress and contouring error inside one constrained optimization.

Full stack review

- Hardware
- Race preparation
- Localization
- Perception
- Planning
- Control

Hardware | Vehicle

1/8-scale RoboRacer platform

- Serpent SRX8GTE R chassis — strong in cornering
- Livox Mid-360 3D LiDAR at 10 Hz (primary sensor) — its built-in IMU also feeds state estimation, so no separate IMU is used



Chassis

Serpent SRX8GTE R 1/8 4WD



Motor

Hobbywing Xerun-4268SD 2400 KV



ESC

VESC 6 MK VI HP



Servo

Highest B2iO HV



Tire

STRADALE SP360 MEGA SOFT



LiDAR

Livox Mid-360 (built-in IMU)

Hardware | Compute platform

Onboard computation runs on an Apple Mac mini (M4)

Our loop is latency-critical and CPU-bound — single-thread throughput, not core count, sets the achievable control rate.

Latency-critical loop

Best-in-class single-thread CPU

KISS-ICP registration and the MPCC solve

share one CPU without contending for memory bandwidth.

Perception headroom

10-core GPU + 16-core ANE · 38 TOPS

The GPU and Neural Engine stay free for on-device perception and learning experiments.

One onboard unit

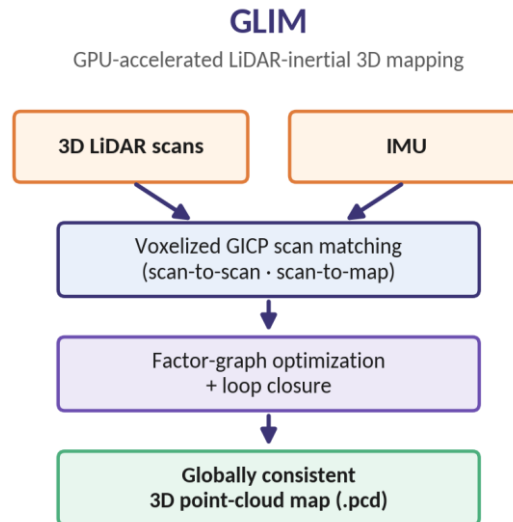
Unified memory · 30–40 W

The whole stack fits in a 30–40 W envelope on a single unit — no extra cooling or power. Replaces the ICRA NUC.

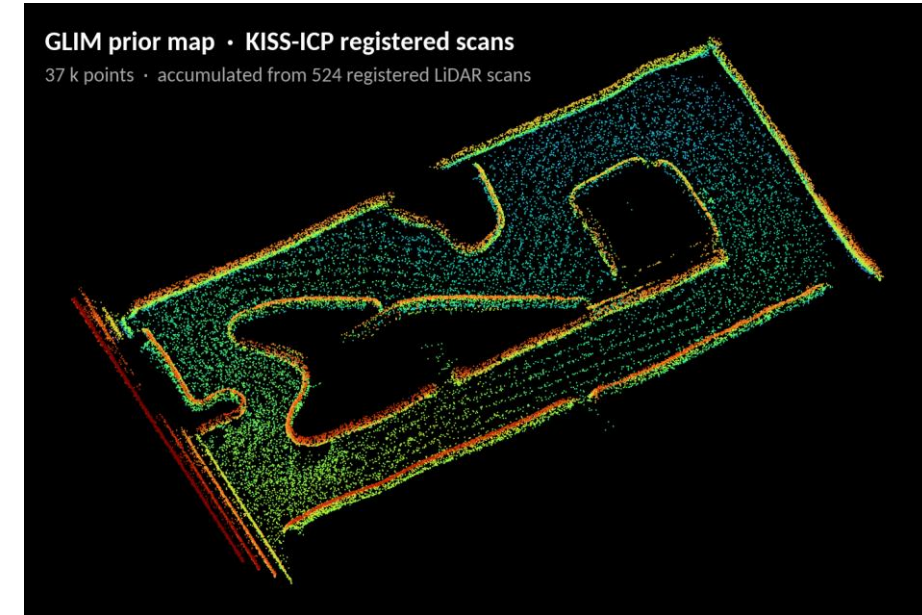
Race preparation | Mapping

Offline mapping with GLIM

- GLIM is a GPU-accelerated LiDAR-inertial SLAM framework: it matches 3D scans with voxelized GICP and refines them in a factor graph with loop closure.
- We run it offline for maximum map fidelity, with no real-time budget to respect.
- Output : a dense, globally consistent 3D point-cloud map (.pcd) — the prior for both localization and raceline optimization.



How GLIM builds the map

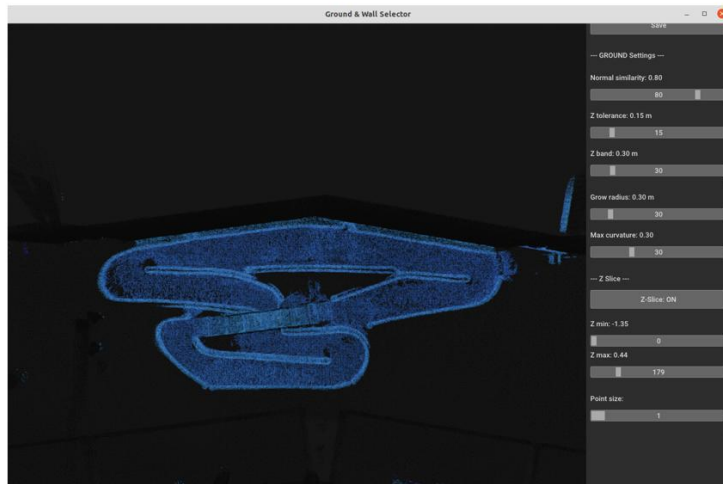


Our track — 524 registered scans

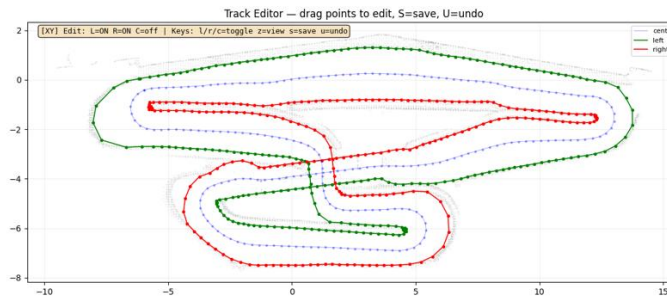
Race preparation | Raceline optimization

Pipeline

- Segment the prior map into ground and walls, extract the centerline and the left / right boundaries, then edit and smooth them
- Iterative minimum-curvature optimization (IQP) → racing line; shortest-path (SP) → complementary reference
- Velocity profile from the combined-acceleration limits (GGV + motor limits)
- Output : dense waypoints — position, heading, curvature, target speed

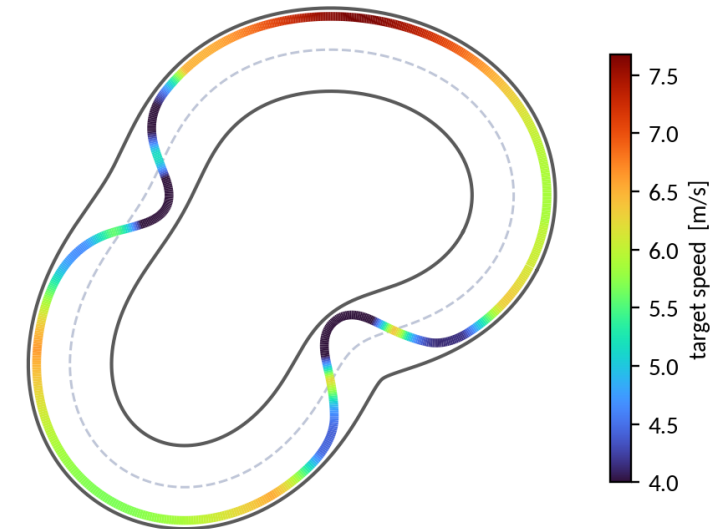


PCD segmentation (ground / wall)



boundary extraction & editing

Optimized racing line with velocity profile



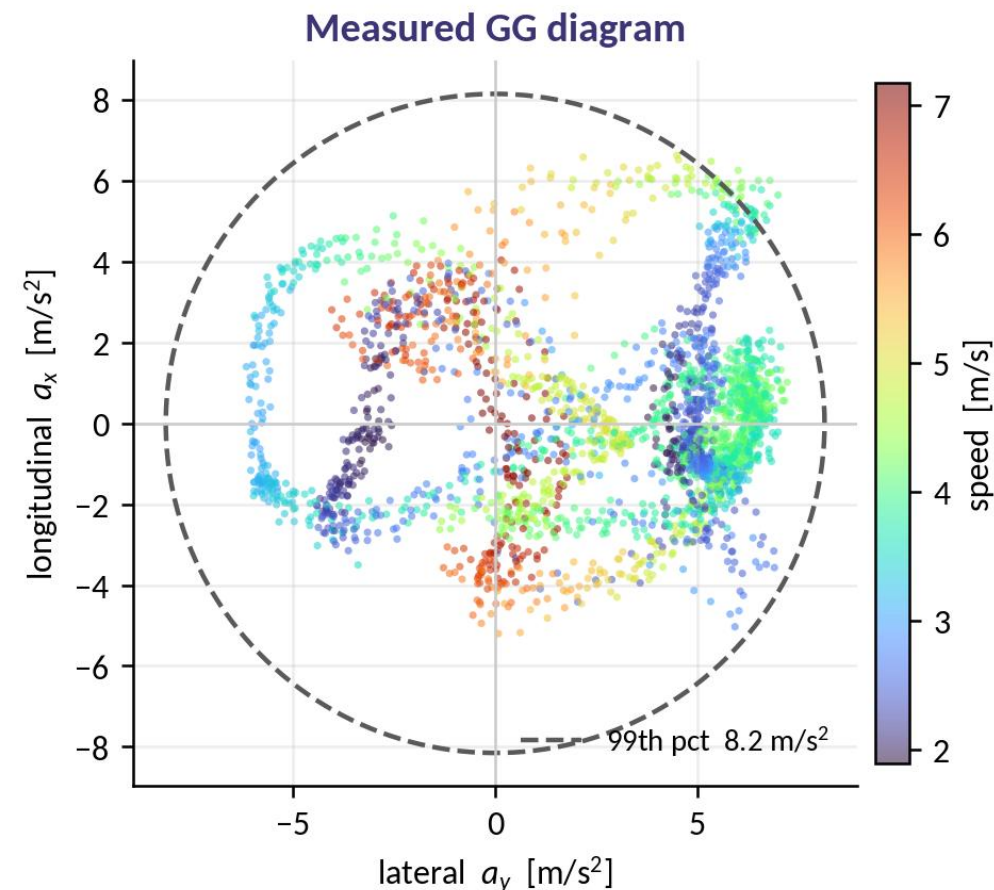
Race preparation | Grip envelope

Combined grip limit

- Longitudinal and lateral grip cannot be demanded at the same time — the achievable envelope shrinks with speed
- Used twice : as the velocity-profile constraint offline, and as the friction-ellipse constraint inside the online MPCC

Measured, not assumed

- The envelope on the right is IMU data from a real run — 8.2 m/s² at the 99th percentile — and it is what we tune the constraint against



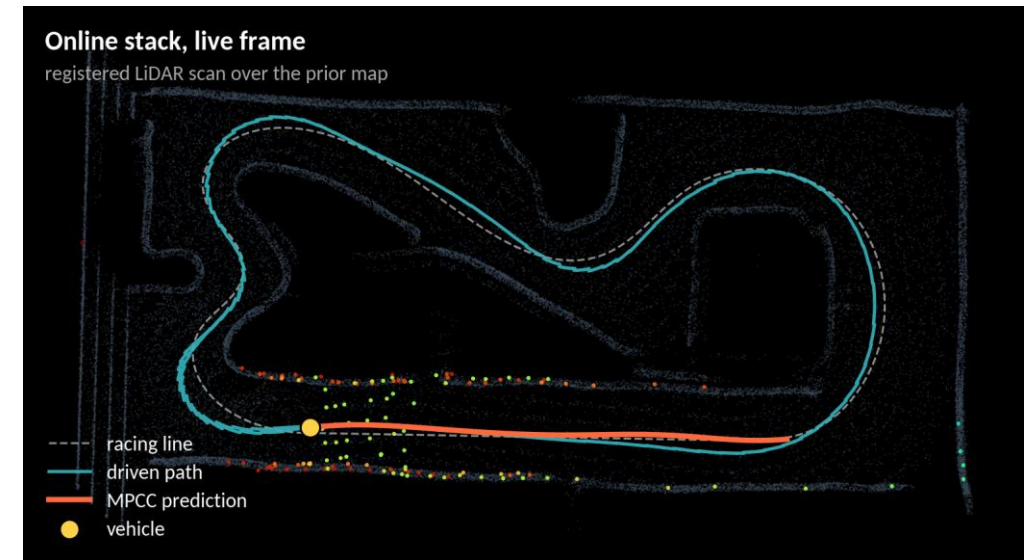
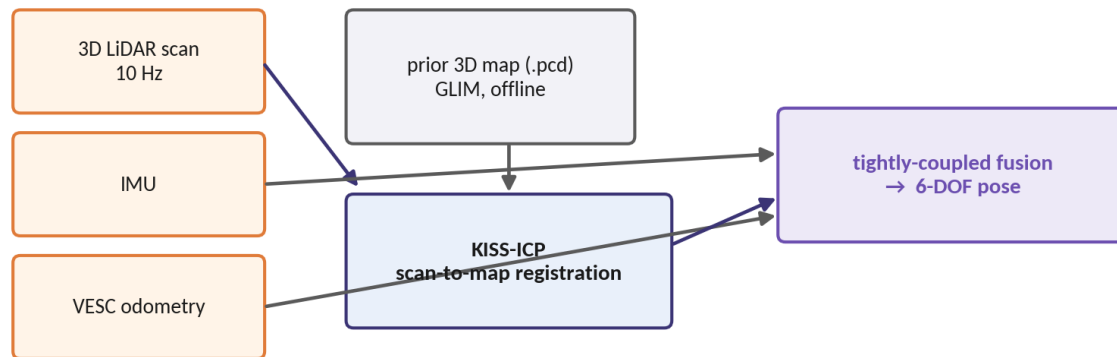
Localization | State estimation

3D LiDAR based state estimation

- Captures the full 6-DOF motion — including the pitch and roll induced by hard acceleration and braking — with dense geometric features for precise pose estimation

Why scan-to-map instead of online SLAM

- The map is a known prior, so registering against it avoids the drift that online mapping accumulates; an ICP-family front-end best exploits a dense offline map
- A single registration front-end keeps the pipeline short and its latency predictable



Perception | Detection & Tracking

Detection pipeline

- Transform to the map frame → subtract the prior map and the ground plane → DBSCAN cluster the residual → Kalman-filter tracking

Why map-aware geometry

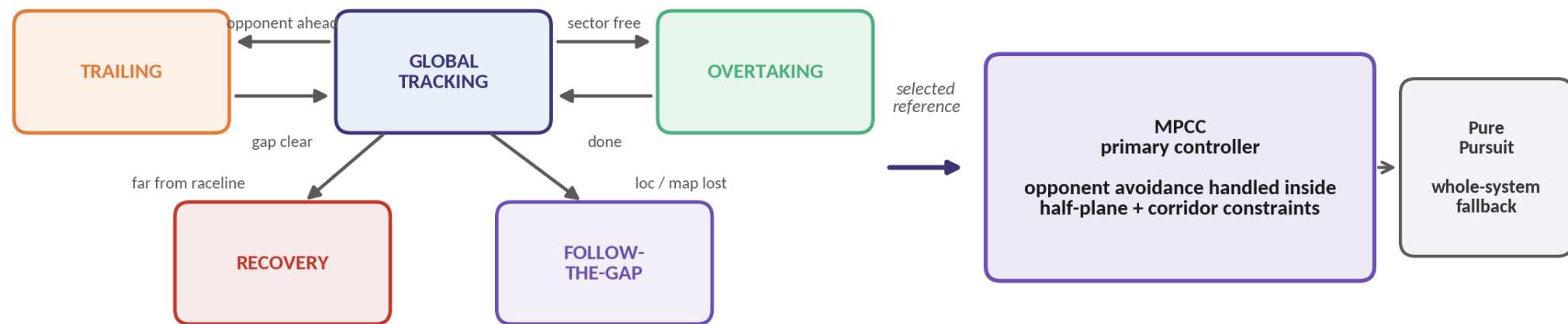
- The map-aware geometric pipeline is deterministic, lightweight, and reuses the prior map.
- FastSAM remains exploratory: it can split one car into wheels and body, making vehicle association ambiguous.



Planning | Local planning

State machine based behavior selection

- Inputs : global raceline, current pose, tracked obstacles → Output : fixed-horizon local waypoints, stitched back onto the raceline
- Transitions are driven by proximity to the raceline, free space ahead, and obstacles in the path
- Trailing locks onto the closest opponent at a safe gap; Overtaking signals the MPCC to pass the opponent — avoidance is handled inside the controller, not by a separate spline; Recovery rejoins from off-track
- Every state feeds the MPCC controller; Pure Pursuit stands by as a whole-system fallback

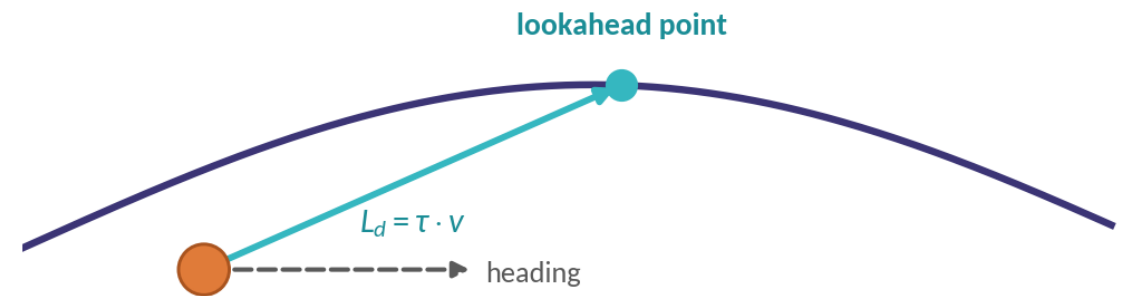


Control | Pure Pursuit baseline

Layer	Enhancement	Why it matters
Lateral	Time-headway lookahead Grip-capped curvature	Speed-aware steering within tire grip
Tracking	Understeer feedforward Heading-error PID	Correct the residual from geometry alone
Longitudinal	Grip-capped raceline speed Alignment gate	Accelerate only when aligned with the path

Role | Robust, low-cost reference baseline; MPCC remains the primary controller.

Pure Pursuit — speed-adaptive lookahead



$$\delta = \arctan(L \kappa), \quad \kappa \leq \kappa_{\text{friction}}$$

Control | Model Predictive Contouring Control

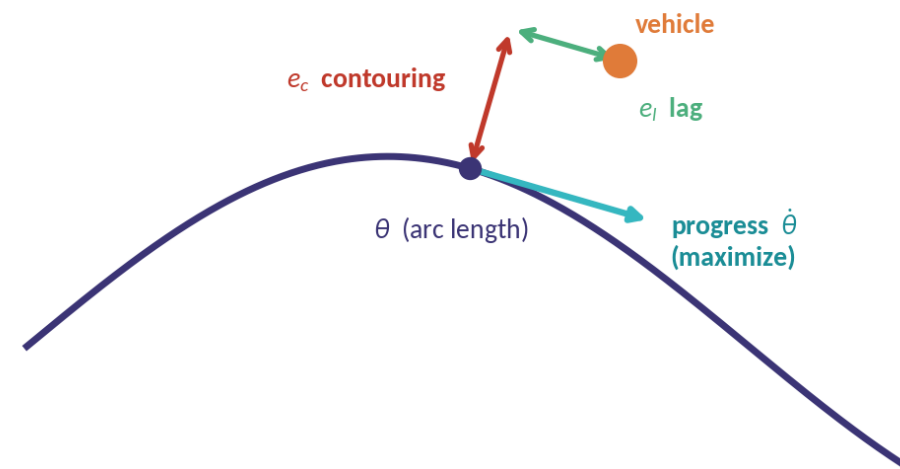
Why MPCC — avoidance comes for free

- Our main motivation: static-obstacle avoidance is naturally handled inside the controller. Each obstacle is just a constraint added to the OCP, so the car passes it without any separate avoidance planner or spline.
- It also trades progress against contouring error in one optimization, and is parameterized by arc length along the racing line — not by time.

Our real-time formulation

- Nonlinear OCP solved every cycle over a short prediction horizon
- Vehicle model blends low-speed kinematic and high-speed dynamic single-track for numerical stability across the full speed range
- Friction-ellipse constraint enforces the tire's combined grip limit

MPCC — contouring / lag error and progress



Control | Real-time solve & avoidance

Solver stack

- acados / HPIPM with a real-time iteration scheme — one SQP iteration per control cycle, so the solve time is bounded and predictable

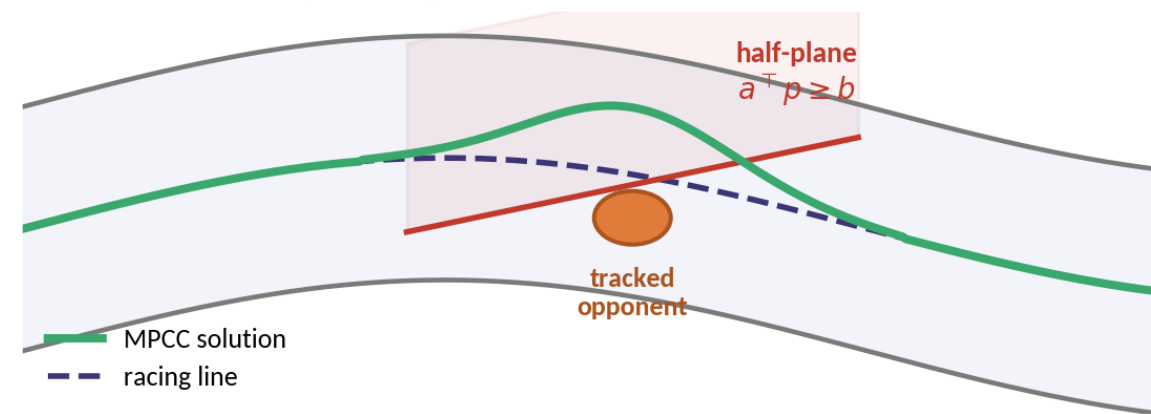
MPCC handles avoidance (primary)

- Tracked obstacles become half-plane constraints and the corridor constraint keeps the trajectory drivable — no separate avoidance planner

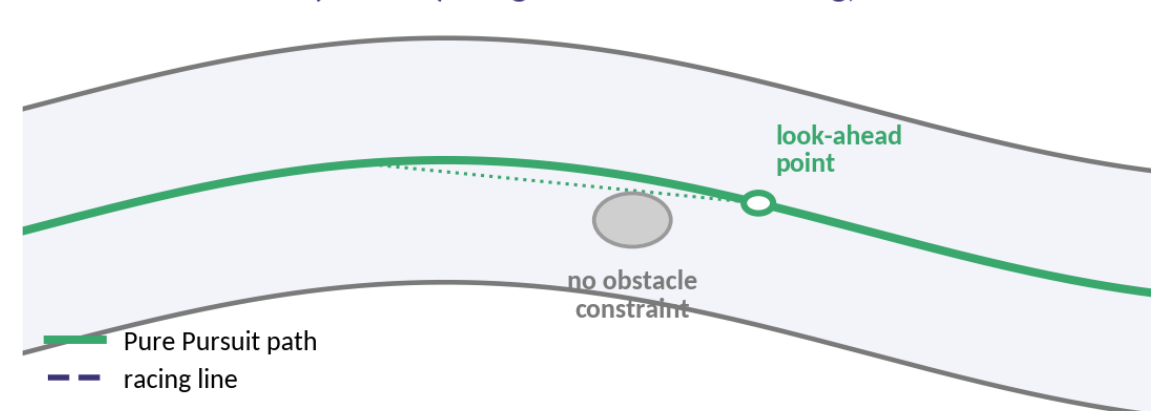
Pure Pursuit is the whole-system fallback

- Engaged only when the MPCC solve is unavailable; pure geometric tracking of the racing line, with no obstacle handling

MPCC (primary) — avoidance solved inside the OCP



Pure Pursuit (fallback) — geometric line tracking, no avoidance



Result

- Stack at a glance

Result | Stack at a glance

OFFLINE PREPARATION

GLIM 3D map

Track boundaries
IQP / SP raceline

GGV target-speed profile

ONLINE EXECUTION

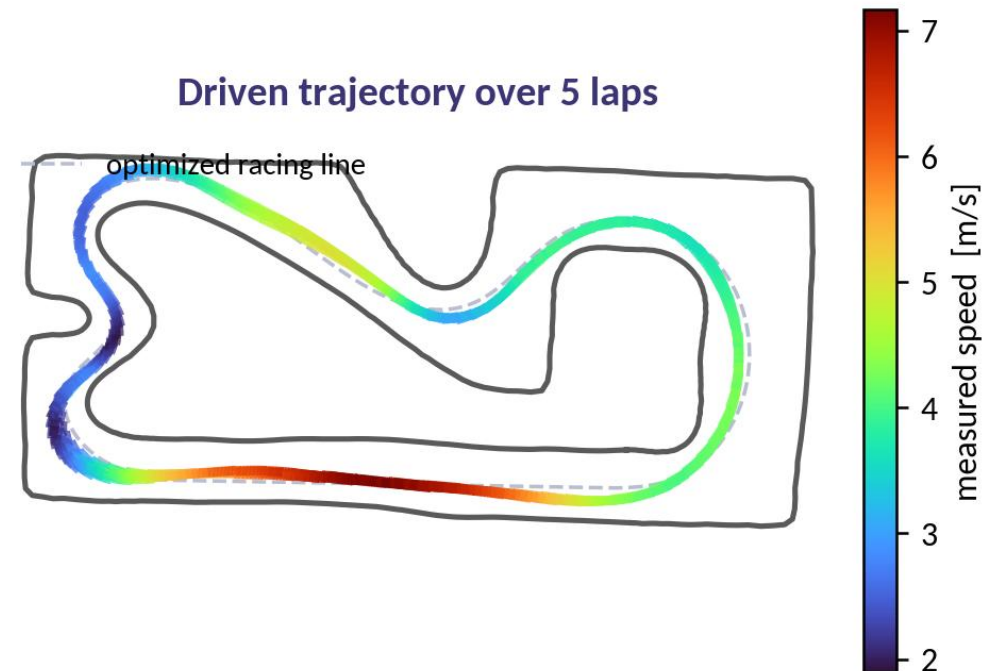
KISS-ICP / IMU / VESC

Map-aware perception
Behavior state

MPCC primary
Pure Pursuit baseline

Half-plane + corridor
constraints inside MPCC

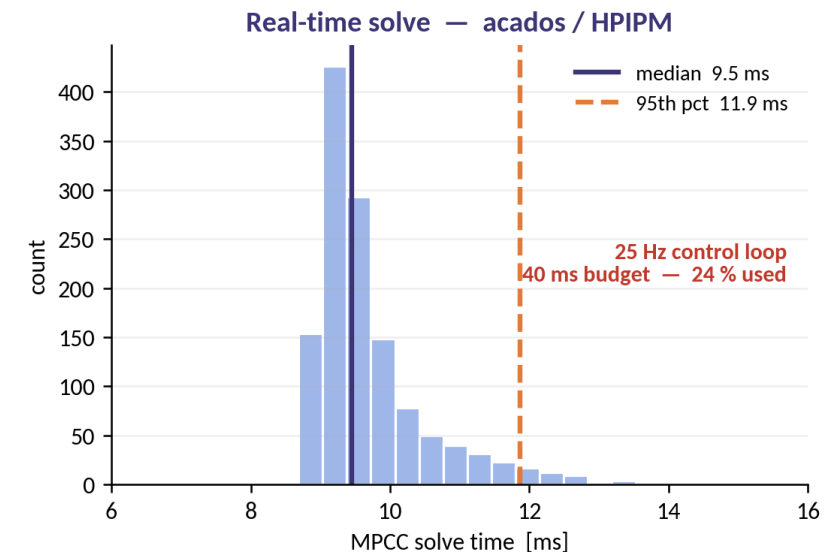
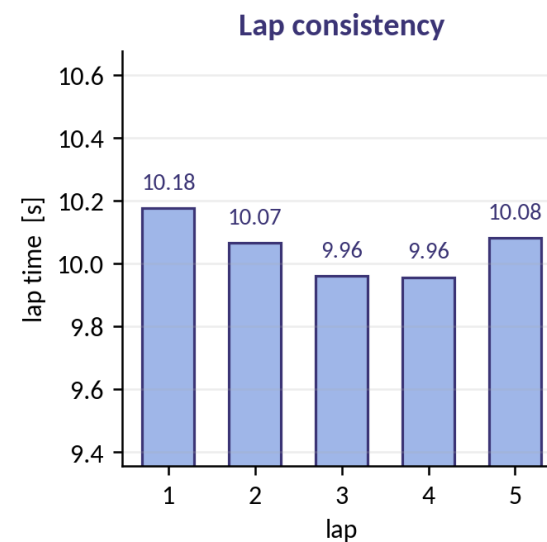
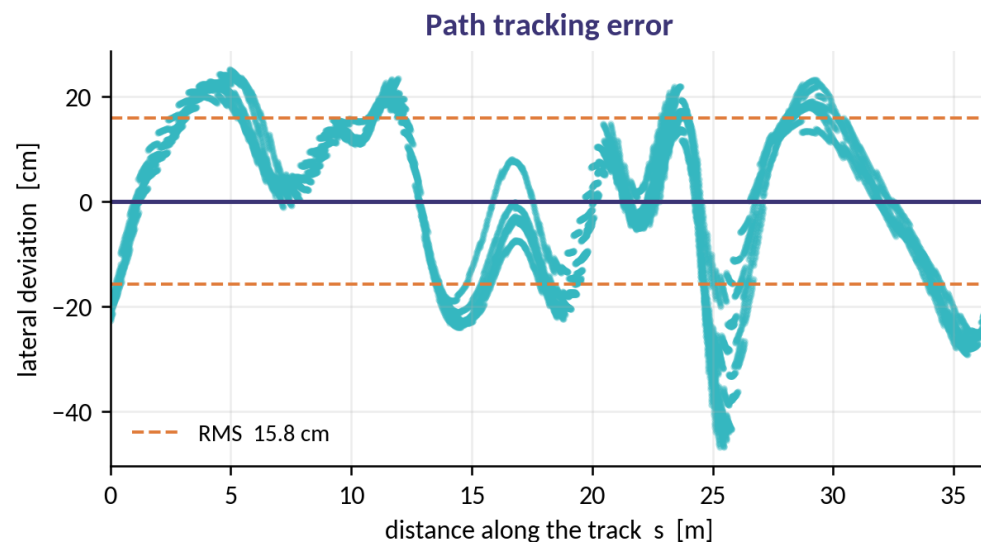
Driven trajectory over 5 laps



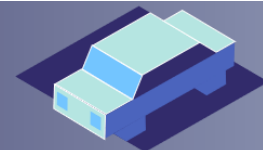
Result | Measured performance

5 laps, 52 s of continuous running

- Best lap 9.96 s · spread 0.22 s over five laps · 36.5 m track
- Top speed 7.2 m/s against a 7.5 m/s planned maximum
- Lateral deviation from the racing line : 15.8 cm RMS
- MPCC solved at 25 Hz — 9.5 ms median, 11.9 ms at the 95th percentile, inside a 40 ms budget; 100 % of solves feasible



THANK YOU



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